Exact vacuum-weighted Wilson-loop variation on a finite interacting lattice

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1 Objects, orientations, and both Lie metrics

Fix an integer $n \geq 2$, a finite oriented graph with vertex set V and one chosen orientation for every physical edge in a nonempty edge set E, and a finite list P of oriented closed plaquette words. Write $N = |\mathsf{E}|$. Reverse traversal of e uses U_e^{-1}, not a new independent link. The configuration manifold and its probability Haar measure are

$$\mathcal{Q} = \mathrm{SU}(n)^{\mathsf{E}}, \qquad d\lambda(U) = \prod_{e \in \mathsf{E}} dU_e, \qquad \lambda(\mathcal{Q}) = 1.$$

For a word $C = ((e_1, \varepsilon_1), \ldots, (e_\ell, \varepsilon_\ell))$ with $\varepsilon_j \in \{1, -1\}$, define, in the written order,

$$H_C(U) = U_{e_1}^{\varepsilon_1} \cdots U_{e_\ell}^{\varepsilon_\ell}, \qquad W_C(U) = \operatorname{tr} H_C(U).$$

A simple edge loop here means a nonempty closed word using each physical edge at most once. Vertices may be repeated; all statements below needing simplicity need precisely the absence of repeated physical edges. Ordinary simple plaquettes and primitive winding loops on a sufficiently large periodic hypercubic lattice satisfy this definition. If a small periodic lattice makes one plaquette repeat an edge, use the repeated-occurrence formula below.

Choose an anti-Hermitian traceless basis T_a, $1 \leq a \leq n^2 - 1$, with

$$-2 \operatorname{tr}(T_a T_b) = \delta_{ab}.$$

The derivatives requested in this calculation are exactly

$$X_{e,a} f(U) = \left. \frac{d}{dt} \right|_{t=0} f(\ldots, e^{tT_a} U_e, \ldots), \qquad \Delta_e^T = \sum_a X_{e,a}^2.$$

In this convention $X_{e,a}$ is skew-adjoint in $L^2(\lambda)$. Indeed, left invariance of Haar gives

$$\int f(e^{tT_a} U_e) g(U_e) dU_e = \int f(U_e) g(e^{-tT_a} U_e) dU_e;$$

differentiation proves the assertion, with the complex conjugate placed on the first argument when appropriate.

There are two retained Lie metrics, related by an explicit dictionary:

$$g_T(A, B) = -2 \operatorname{tr}(AB), \qquad c(A, B) = -\tfrac{1}{2} \operatorname{tr}(AB) = \tfrac{1}{4} g_T(A, B), \tag{1}$$

$$c(T_a, T_b) = \tfrac{1}{4} \delta_{ab}, \qquad S_a = 2T_a, \qquad c(S_a, S_b) = \delta_{ab},$$

$$X_{e,a}^c = 2X_{e,a}, \qquad \Delta_e^c = 4\Delta_e^T. \tag{2}$$

Thus a Casimir eigenvalue measured with c is four times its g_T value. The Hamiltonian in this paper is the specified operator

$$H = -\sum_{e\in\mathsf{E}} \kappa_e \Delta_e^T + V, \quad V(U) = \sum_{p\in\mathsf{P}} b_p\big(n - \operatorname{Re} W_p(U)\big), \quad \kappa_e > 0,\ b_p > 0. \tag{3}$$

In the original cusp metric it is identically $H = -\sum_e (\kappa_e/4)\Delta_e^c + V$. Keeping the coefficient κ_e in front of Δ_e^c would instead multiply the kinetic operator by four and would define a different Hamiltonian. No such replacement is made here. The inequalities $|\operatorname{tr} U| \le n$ for $U \in \mathrm{SU}(n)$ and

$$0 \le V \le M, \qquad M = 2nB, \qquad B = \sum_{p\in\mathsf{P}} b_p, \tag{4}$$

retain every plaquette term. The trace bound follows by summing the n unit-modulus eigenvalues. The constant $2nB$ is a valid bound; no assertion that all plaquettes attain their separate maxima simultaneously is needed.

The probability Haar convention also has an exact volume dictionary. On the compact configuration manifold, whose real dimension is $d = N(n^2-1)$, (1) gives $d\operatorname{vol}_c = 2^{-d} d\operatorname{vol}_{g_T}$. If $\mathcal{V}_c = \operatorname{vol}_c(\mathcal{Q})$ and $\mathcal{V}_T = \operatorname{vol}_{g_T}(\mathcal{Q})$, then

$$d\lambda = \mathcal{V}_c^{-1} d\operatorname{vol}_c = \mathcal{V}_T^{-1} d\operatorname{vol}_{g_T}, \qquad \mathcal{V}_c = 2^{-d}\mathcal{V}_T.$$

The map $f \mapsto \sqrt{\mathcal{V}_c}\, f$ is unitary from $L^2(d\operatorname{vol}_c)$ to $L^2(d\lambda)$ and intertwines the same differential Hamiltonian. A ground state normalized in the former space is therefore taken to $\psi_0 = \sqrt{\mathcal{V}_c}\,\psi_c$, so $\psi_0^2 d\lambda = \psi_c^2 d\operatorname{vol}_c$ exactly. These are volume identities on the compact group product only; they do not identify a noncompact cusp measure with Haar measure.

Gauge transformations are $(g\cdot U)_e = g_{s(e)} U_e g_{t(e)}^{-1}$ for $g \in \mathrm{SU}(n)^{\mathsf{V}}$. Along a closed word all adjacent factors cancel, so H_C changes by conjugation at its base vertex and W_C is invariant. Haar measure, each Δ_e^T, and V are gauge invariant. The physically restricted finite Hilbert space is therefore the closed invariant subspace $\mathcal{H}_{\mathrm{inv}} \subset L^2(\mathcal{Q}, \lambda)$, and H preserves it.

2 Fierz identities with the stated normalization

Lemma 1. *For arbitrary complex $n \times n$ matrices A, B,*

$$\sum_a (T_a)_{ij}(T_a)_{kl} = -\frac{1}{2}\left(\delta_{il}\delta_{jk} - \frac{1}{n}\delta_{ij}\delta_{kl}\right), \tag{5}$$

$$\sum_a \operatorname{tr}(T_a A)\operatorname{tr}(T_a B) = -\frac{1}{2}\left(\operatorname{tr}(AB) - \frac{1}{n}\operatorname{tr} A \operatorname{tr} B\right), \tag{6}$$

$$\sum_a \operatorname{tr}(T_a A)\overline{\operatorname{tr}(T_a B)} = \frac{1}{2}\left(\operatorname{tr}(AB^\dagger) - \frac{1}{n}\operatorname{tr} A\,\overline{\operatorname{tr} B}\right), \tag{7}$$

$$\sum_a T_a^2 = -C_F I_n, \qquad C_F = \frac{n^2-1}{2n}. \tag{8}$$

In particular, for $A \in \mathrm{SU}(n)$,

$$\sum_a |\operatorname{tr}(T_a A)|^2 = \frac{n^2 - |\operatorname{tr} A|^2}{2n}. \tag{9}$$

Proof. The matrices $Q_0 = I_n/\sqrt{n}$ and $Q_a = i\sqrt{2}T_a$ form an orthonormal basis of the real vector space of Hermitian matrices for the scalar product $\operatorname{tr}(QR)$. Their complex linear span is all complex matrices. Expanding an arbitrary matrix D gives $D = \sum_{\alpha=0}^{n^2-1} Q_\alpha \operatorname{tr}(Q_\alpha D)$. Taking the coefficient of D_{lk} in entry ij proves $\sum_\alpha (Q_\alpha)_{ij}(Q_\alpha)_{kl} = \delta_{il}\delta_{jk}$. Subtract the Q_0 term and use $(i\sqrt{2})^2 = -2$ to obtain (5). Contract it with $A_{ji}B_{lk}$ to get (6). Anti-Hermiticity gives $\overline{\operatorname{tr}(T_aB)} = -\operatorname{tr}(T_aB^\dagger)$; this proves (7). Put $k = j$ in (5) and sum j to obtain (8). Finally $AA^\dagger = I_n$ gives (9). □

3 Every edge orientation and the exact overlap morphism

Fix one occurrence of physical edge e in C and write the ordered word as $PU_e^\varepsilon Q$, where P and Q are the products before and after that occurrence. Cyclic trace invariance gives the following matrices based at the original source $s(e)$:

$$A_{C,e,r} = \begin{cases} U_e QP, & \varepsilon = +1, \\ QPU_e^{-1}, & \varepsilon = -1. \end{cases} \tag{10}$$

The occurrence index r is suppressed when there is only one occurrence. Each $A_{C,e,r}$ lies in $\mathrm{SU}(n)$, has trace W_C, and transforms by conjugation with $g_{s(e)}$ under a gauge transformation. These statements follow either by multiplying the factors or by following the cyclically based closed path at $s(e)$.

The derivative of an inverse is $X_{e,a}U_e^{-1} = -U_e^{-1}T_a$, since $(e^{tT_a}U_e)^{-1} = U_e^{-1}e^{-tT_a}$. Consequently

$$\begin{aligned} X_{e,a}\operatorname{tr}(PU_eQ) &= \operatorname{tr}(PT_aU_eQ) = \operatorname{tr}(T_aU_eQP), \\ X_{e,a}\operatorname{tr}(PU_e^{-1}Q) &= -\operatorname{tr}(PU_e^{-1}T_aQ) = -\operatorname{tr}(T_aQPU_e^{-1}). \end{aligned} \tag{11}$$

For an arbitrary closed word the product rule now gives

$$X_{e,a}W_C = \sum_{r:e_r=e} \varepsilon_r \operatorname{tr}(T_aA_{C,e,r}). \tag{12}$$

All factors in (10) are evaluated at the original configuration; there is no independent-link approximation in this formula.

For two words define the edgewise sesquilinear and bilinear contractions by

$$\Gamma_e(F,G) = \sum_a (X_{e,a}F)\overline{X_{e,a}G}, \qquad \Gamma_e^{\mathrm{bil}}(F,G) = \sum_a (X_{e,a}F)(X_{e,a}G).$$

Lemma 1 proves, occurrence by occurrence,

$$\Gamma_e(W_C,W_D) = \frac{1}{2}\sum_{r:e_r=e}\sum_{s:d_s=e} \varepsilon_r\eta_s \left[\operatorname{tr}(A_{C,e,r}A_{D,e,s}^\dagger) - \frac{1}{n}W_C\overline{W_D}\right], \tag{13}$$

$$\Gamma_e^{\mathrm{bil}}(W_C,W_D) = -\frac{1}{2}\sum_{r:e_r=e}\sum_{s:d_s=e} \varepsilon_r\eta_s \left[\operatorname{tr}(A_{C,e,r}A_{D,e,s}) - \frac{1}{n}W_CW_D\right]. \tag{14}$$

These identities construct the exact relation between the two loop variations: $A_{C,e,r}A_{D,e,s}^{-1}$ is the concatenation of the first based loop and the inverse of the second, and its trace is gauge invariant. The nonconjugated contraction concatenates the two based loops directly. An edge not occurring in both words contributes zero.

For simple edge loops, a common edge contributes exactly

$$\Gamma_e(W_C, W_D) = \frac{\varepsilon_{Ce}\varepsilon_{De}}{2}\left[\operatorname{tr}(A_{Ce}A_{De}^{\dagger}) - \frac{1}{n}W_C\overline{W_D}\right], \tag{15}$$
$$\Gamma_e^{\mathrm{bil}}(W_C, W_D) = -\frac{\varepsilon_{Ce}\varepsilon_{De}}{2}\left[\operatorname{tr}(A_{Ce}A_{De}) - \frac{1}{n}W_C W_D\right].$$

In particular, for every used edge of a simple loop,

$$\Gamma_e(W_C, W_C) = \frac{n^2 - |W_C|^2}{2n}, \qquad \sum_a X_{e,a}^2 W_C = -C_F W_C. \tag{16}$$

The second equality follows directly from (11) and (8); in negative orientation the two derivative signs cancel. Both equalities are zero on unused edges. Define $K_C = \sum_{e\in C}\kappa_e$. Then the full pointwise energy is

$$\Gamma_\kappa(W_C, W_C) := \sum_e \kappa_e\Gamma_e(W_C, W_C) = \frac{K_C}{2n}(n^2 - |W_C|^2). \tag{17}$$

For later real plaquette observables the exact companion identity is

$$\Gamma_\kappa(\operatorname{Re} W_C, \operatorname{Re} W_C) = \frac{K_C}{4}\left[n - \operatorname{Re}\operatorname{tr}(H_C^2) - \frac{2}{n}(\operatorname{Im} W_C)^2\right]. \tag{18}$$

To prove it, use $(\operatorname{Re} z)^2 = (|z|^2 + \operatorname{Re} z^2)/2$ in the derivative sum, apply both contractions in Lemma 1, and observe $\operatorname{tr}(A_{Ce}^2) = \operatorname{tr}(H_C^2)$ because cyclic rotations of invertible factors are conjugate matrices. Thus the right side is nonnegative despite its written difference of terms. The same energy holds for $n - \operatorname{Re} W_C$. Measured using $X^c = 2X$, every Γ_e in this section is multiplied by four; pairing it with kinetic coefficient $\kappa_e/4$ recovers exactly (17) and (18).

4 Existence, exact ground-state transform, and orthogonality

The domain of the nonnegative quadratic form of H is the first Sobolev space on $\mathcal{Q}$, with the equivalent norm given by the finitely many fields $X_{e,a}$. Its form is

$$q_H(u, v) = \sum_{e,a}\kappa_e \int \overline{X_{e,a}u}X_{e,a}v\, d\lambda + \int V\overline{u}v\, d\lambda.$$

We use the self-adjoint operator associated to this closed form.

Lemma 2. *H has a unique normalized strictly positive ground eigenfunction $\psi_0 \in C^\infty(\mathcal{Q})$, with $H\psi_0 = E_0\psi_0$ and $\int \psi_0^2 d\lambda = 1$. It is gauge invariant. In particular the bottom eigenvalue of the gauge-invariant restriction is the same E_0.*

Proof. The free positive operator $H_0 = -\sum_e \kappa_e\Delta_e^T$ has a complete matrix-coefficient eigenbasis by the compact-group Peter–Weyl theorem [1]; its finite-multiplicity eigenvalues tend to infinity. The latter fact also follows from the explicit weights and bounds proved in Section 6. Its resolvent is compact. Bounded real multiplication by V preserves self-adjointness and compact resolvent: for sufficiently negative z, the identity $(H - z)^{-1} = (H_0 - z)^{-1}[I + V(H_0 - z)^{-1}]^{-1}$ proves compactness by a convergent Neumann series. Hence the bottom of the spectrum is an eigenvalue. The inequality

$|X|u|| \le |Xu|$ almost everywhere (obtain it first for $\sqrt{|u|^2+\epsilon^2}$ and then let $\epsilon \downarrow 0$) shows that the modulus of a minimizing eigenfunction is also minimizing. Choose a nonnegative normalized one. The elliptic equation $H_0\psi_0 = (E_0 - V)\psi_0$ and local elliptic regularity, iterated with smooth V, give $\psi_0 \in C^\infty$.

For completeness, positivity needs no assumption about an unknown vacuum. The parabolic comparison inequality proved in Section 6 is $e^{-tH} f \ge e^{-Mt} e^{-tH_0} f$ for $f \ge 0$. At the explicit time t_* in that section the kernel of $e^{-t_* H_0}$ is at least 2^{-N}. Thus $\psi_0(U) \ge e^{(E_0 - M)t_*} 2^{-N} \int \psi_0 d\lambda > 0$. The next theorem, whose calculation uses only this already established positivity and the eigenvalue equation, proves uniqueness: if u is another ground eigenfunction, put $f = u/\psi_0$ in its identity. The result is $\int \psi_0^2 \sum_{e,a} \kappa_e |X_{e,a} f|^2 d\lambda = 0$. The fields span each tangent space and $\mathcal{Q}$ is connected, so f is constant. Gauge transformations preserve H, positivity and normalization, and consequently fix ψ_0 by this uniqueness. □

Set

$$d\mu = \rho \, d\lambda, \qquad \rho = \psi_0^2, \qquad \mu(F) = \int F \, d\mu. \tag{19}$$

This measure is determined by the interacting Hamiltonian (3); no other measure is substituted for it below. Multiplication $\mathcal{U} f = \psi_0 f$ is a unitary map from $L^2(\mu)$ onto $L^2(\lambda)$, since $\|\mathcal{U} f\|_\lambda^2 = \int |f|^2 d\mu$, and division by the positive ψ_0 is its inverse. It also maps the respective gauge-invariant subspaces onto one another.

Theorem 3. *For all smooth complex f, g,*

$$\begin{aligned} Lg := \mathcal{U}^{-1}(H - E_0)\mathcal{U} g &= -\sum_{e,a} \kappa_e \big[X_{e,a}^2 g + 2(X_{e,a} \log \psi_0) X_{e,a} g \big] \\ &= -\rho^{-1} \sum_{e,a} \kappa_e X_{e,a}(\rho X_{e,a} g), \end{aligned} \tag{20}$$

$$\langle \psi_0 f, (H - E_0)\psi_0 g \rangle_\lambda = \sum_{e,a} \kappa_e \int \overline{X_{e,a} f} X_{e,a} g \, d\mu. \tag{21}$$

The identities extend to the corresponding closed form domains. For any nonconstant smooth F, the vector $u_F = \psi_0(F - \mu(F))$ is orthogonal to ψ_0, and

$$\frac{\langle u_F, (H - E_0) u_F \rangle}{\|u_F\|^2} = \frac{\mu(\Gamma_\kappa(F,F))}{\mu(|F|^2) - |\mu(F)|^2}. \tag{22}$$

Proof. The product rule gives $X^2(\psi_0 g) = \psi_0 X^2 g + 2(X\psi_0) X g + g X^2 \psi_0$ for each X. Subtracting $g(H - E_0)\psi_0 = 0$ from $(H - E_0)(\psi_0 g)$ proves (20); its divergence form uses $X\rho = 2\psi_0 X\psi_0$. Multiply by $\rho \overline{f}$ and integrate. The skew-adjointness of X with respect to λ, proved above, gives $-\int \overline{f} X(\rho X g) d\lambda = \int \rho \overline{X f} X g d\lambda$. Summing proves (21). On a compact manifold multiplication and division by the smooth positive ψ_0 are bounded on the first Sobolev space, so smooth approximation proves the form extension. Finally $\langle \psi_0, u_F \rangle = \mu(F) - \mu(F) = 0$, $\|u_F\|^2 = \mu(|F|^2) - |\mu(F)|^2$, and subtracting a constant leaves all derivatives unchanged. The variance is strictly positive because F is nonconstant and $\rho > 0$ everywhere. These observations prove (22). □

In particular, put $m_C = \mu(W_C)$ and $r_C = \mu(|W_C|^2)$. For a simple edge loop the exact answer is

$$Q_C = \frac{K_C}{2n} \, \frac{n^2 - r_C}{r_C - |m_C|^2}. \tag{23}$$

The potential disappears from the displayed Dirichlet integrand because the exact ground-state equation was used. It remains in ρ, m_C, and r_C, as well as in the drift term in (20). Thus this identity does not set V or any b_p to zero.

5 Center symmetry, exact selection rules, and character limits

Assign a central element $z_e \in Z(\mathrm{SU}(n))$ to each physical edge, subject to

$$\prod_{r\in p} z_{e_r}^{\varepsilon_r} = 1 \quad \text{for every retained plaquette } p. \tag{24}$$

Define $\sigma_z(U)_e = z_e U_e$. This is a Haar-preserving diffeomorphism, commutes with the edge derivatives, and preserves every plaquette holonomy because the factors z_e are central and obey (24). It therefore preserves H and, by uniqueness and positivity, ψ_0 and μ. For every closed word,

$$W_C(\sigma_z U) = \zeta_C W_C(U), \qquad \zeta_C = \prod_{r\in C} z_{e_r}^{\varepsilon_r}.$$

Integration proves the exact selection rules

$$\mu(W_C) = 0 \quad \text{if } \zeta_C \neq 1, \qquad \mu(W_C\overline{W_D}) = 0 \quad \text{if } \zeta_C\overline{\zeta_D} \neq 1. \tag{25}$$

For powers and products the same proof applies with their product charge.

On a periodic hypercubic lattice, multiply the links in one coordinate direction crossing one fixed periodic cut by $z \in Z(\mathrm{SU}(n))$, and leave the other links unchanged. Each oriented plaquette crosses that cut with total signed count zero, which verifies (24) directly. A loop with integer winding q in that direction acquires z^q. A primitive single winding has a nontrivial charge for a nonidentity suitable z, and therefore $m_C = 0$. A loop whose winding in every direction is divisible by n does not obtain a zero mean from this argument. Contractible plaquettes are center neutral; their means must be retained.

For a charged simple loop the quotient becomes

$$Q_C = \frac{K_C}{2n}\left(\frac{n^2}{r_C} - 1\right), \qquad 0 < r_C < n^2. \tag{26}$$

The strict inequalities follow from $\rho > 0$ and the fact that the holonomy map of a simple loop has image all of $\mathrm{SU}(n)$: fix the other edges and vary any one used edge. Some images have trace zero and others have trace n, and neighborhoods of both have positive Haar measure. In particular center symmetry does not set r_C equal to one.

Lemma 4. *For every simple edge loop,*

$$\int W_C d\lambda = 0, \quad \int |W_C|^2 d\lambda = 1, \quad \int (n^2 - |W_C|^2) d\lambda = n^2 - 1.$$

Proof. Condition on every edge except one appearing once. Its loop holonomy is $AU_e^{\pm 1}B$ for fixed group matrices A, B, and is Haar distributed by left and right invariance and invariance under inversion. It suffices to integrate over one Haar matrix U. Its mean trace is zero since $U \mapsto zU$ preserves Haar for a nonidentity central z. For arbitrary D, the matrix $\int UDU^\dagger dU$ commutes with every $\mathrm{SU}(n)$ matrix by invariance, and so is $(\operatorname{tr} D)I_n/n$. To see the scalar conclusion without an additional representation assumption, commuting with diagonal unitary matrices kills the off-diagonal entries and commuting with determinant-one unitary changes of coordinate axes makes all diagonal entries equal. Set $D = E_{jl}$ and compare entries to get $\int U_{ij}\overline{U_{kl}} dU = \delta_{ik}\delta_{jl}/n$. Taking $j = i$, $l = k$ and summing i, k gives $\int |\operatorname{tr} U|^2 dU = 1$. The last asserted integral follows by subtraction. □

There is also an exact character map that retains the second moment. The conjugation action $A \mapsto UAU^{-1}$ on $\mathrm{End}(\mathbb{C}^n)$ has trace $\operatorname{tr} U \operatorname{tr} U^{-1} = |\operatorname{tr} U|^2$. The invariant decomposition into scalar and traceless matrices has characters 1 and χ_{ad}. Consequently

$$|\operatorname{tr} U|^2 = 1 + \chi_{\mathrm{ad}}(U), \quad r_C = 1 + \mu(\chi_{\mathrm{ad}}(H_C)), \quad -1 \le \chi_{\mathrm{ad}}(U) \le n^2 - 1. \tag{27}$$

This derivation proves the equality and both bounds directly; the adjoint character is center neutral, so (25) imposes no zero expectation on it.

Proposition 5. *Center and gauge invariance, positivity of a smooth density, and the pointwise character bounds alone imply no strictly positive uniform lower bound on r_C and no upper bound strictly smaller than n^2. This assertion concerns those premises alone, not the narrower family of ground states of* (3)*.*

Proof. Let $h(U) = |W_C(U)|^2$. Its range includes 0 and n^2. For an explicit trace-zero matrix take $\zeta = e^{2\pi i/n}$ and $A = \alpha \operatorname{diag}(1, \zeta, \ldots, \zeta^{n-1})$ with $\alpha = e^{-\pi i(n-1)/n}$. The geometric sum is zero and $\det A = \alpha^n \zeta^{n(n-1)/2} = 1$. Choose one edge to realize A as holonomy; identity holonomy realizes n^2. For each $s > 0$, the densities

$$\rho_s^- = \frac{e^{-sh}}{\int e^{-sh} d\lambda}, \qquad \rho_s^+ = \frac{e^{sh}}{\int e^{sh} d\lambda}$$

are smooth, strictly positive, gauge invariant, and invariant under every center transformation of the stated type. For $\epsilon > 0$, the set $h < \epsilon/2$ is nonempty open and has Haar measure $a_\epsilon > 0$. Its contribution to the first denominator is at least $a_\epsilon e^{-s\epsilon/2}$, whereas the numerator mass of $h \ge \epsilon$ is at most $e^{-s\epsilon}$. Thus this latter probability tends to zero and $\int h \rho_s^- d\lambda \to 0$ by boundedness. Apply the same argument to $n^2 - h$ for ρ_s^+ to obtain $\int h \rho_s^+ d\lambda \to n^2$. For a nontrivially charged loop, both measures also have zero loop mean by the selection-rule proof. No claim is made that these constructed densities arise from positive Wilson plaquette coefficients in (3); their role is to prove exactly the asserted insufficiency of symmetry and character bounds alone. □

6 Explicit all-coupling finite-volume estimates

The full Hamiltonian supplies more information than those symmetry premises. The following bounds use it and hold for arbitrary positive values of all retained couplings. Define

$$\kappa_* = \min_{e \in \mathsf{E}} \kappa_e, \quad D = n^2 - 2, \quad t_* = \frac{2(D \log 2 + \log 3)}{(n-1)\kappa_*}, \quad a_* = 3^{-2N} e^{-2Mt_*} = 3^{-2N} e^{-4nBt_*}. \tag{28}$$

These constants are dimensionally consistent: t_* is inverse energy, M is energy, and a_* is dimensionless. They are estimates, not a claim of optimal constants.

Lemma 6. *The integral kernel of $P_t = e^{-tH_0}$ satisfies*

$$2^{-N} \le p_{t_*}(U, V) \le (3/2)^N \qquad \text{for all } U, V \in \mathcal{Q}. \tag{29}$$

Proof. We give the representation constants to fix the heat time and Lie metric exactly. The irreducibles of $\mathrm{SU}(n)$ are indexed by $m = (m_1, \ldots, m_{n-1}) \in \mathbb{Z}_{\ge 0}^{n-1}$, with $\lambda_m = \sum_i m_i \omega_i$. In the usual Euclidean root space $\{x \in \mathbb{R}^n : \sum x_i = 0\}$ with roots $e_i - e_j$ of squared length two, set

$$\omega_i = \sum_{j=1}^{i} e_j - \frac{i}{n} \sum_{j=1}^{n} e_j, \quad \rho_{\mathrm{rt}} = \sum_{i=1}^{n-1} \omega_i.$$

The dimension and the g_T-Casimir eigenvalue are

$$d_m = \prod_{1\le i<j\le n} \frac{\sum_{q=i}^{j-1} m_q + j - i}{j-i}, \tag{30}$$

$$C_m = \frac{1}{2}(\lambda_m, \lambda_m + 2\rho_{\mathrm{rt}}). \tag{31}$$

These standard representation identities can be read in row coordinates in [2]. To spell out the dictionary and constants, put $\ell_i = \sum_{q=i}^{n-1} m_q$, $\ell_n = 0$, and $L = \sum_i \ell_i$. The highest weight has coordinates $\ell_i - L/n$, and $2\rho_{\mathrm{rt},i} = n + 1 - 2i$. Thus (31) says explicitly

$$C_m = \frac{1}{2}\left[\sum_{i=1}^{n} \ell_i(\ell_i + n + 1 - 2i) - \frac{L^2}{n}\right].$$

The dimension formula follows from the character alternant $\det(z_i^{\ell_j+n-j})/\det(z_i^{n-j})$ by letting all z_i tend to one: the first nonzero alternant terms are the two Vandermonde determinants in $\ell_j + n - j$ and $n - j$, whose ratio is (30). The Casimir formula follows by applying the Cartan and root decomposition of the quadratic Casimir to a highest weight vector: the Cartan term gives (λ_m, λ_m), and commuting each positive-root raising operator past its lowering operator gives $2(\lambda_m, \rho_{\mathrm{rt}})$; the factor $1/2$ is fixed by (8). Hence the metric normalization is the one used in (3), rather than an unspecified heat-kernel convention.

Write $k = \sum_i m_i$. Direct scalar multiplication gives $(\omega_i, \rho_{\mathrm{rt}}) = i(n-i)/2$. Since $(\lambda_m, \lambda_m) \ge 0$ and $i(n-i) \ge n - 1$,

$$C_m \ge \frac{n-1}{2}k, \qquad d_m \le (k+1)^{n(n-1)/2}.$$

The number of m with this sum k is $\binom{k+n-2}{n-2} \le (k+1)^{n-2}$: the first $n-2$ entries determine the last, and each has at most $k+1$ choices. This also covers $n = 2$.

Peter–Weyl orthogonality and the Casimir action give the single-edge heat kernel

$$q_s(g) = \sum_m d_m e^{-sC_m} \chi_m(g).$$

Indeed convolution by $d_m\chi_m$ projects onto the matrix coefficients of the corresponding irreducible (with the inverse in the kernel argument if needed), and the heat operator multiplies that subspace by e^{-sC_m}. The series is absolutely uniformly convergent for every $s > 0$: a unitary representation has $|\chi_m(g)| \le d_m$, and the above bounds majorize the nonconstant terms by a polynomial in k times $e^{-s(n-1)k/2}$. This also proves finite eigenvalue multiplicities and divergence of the eigenvalues used in Lemma 2. The constant representation contributes one. For edge e and time t,

$$\begin{aligned} |q_{\kappa_e t}(g) - 1| &\le \sum_{k\ge 1} (k+1)^D e^{-\kappa_e t(n-1)k/2} \\ &\le \sum_{k\ge 1} \left(2^D e^{-\kappa_e t(n-1)/2}\right)^k. \end{aligned}$$

The second inequality uses $k + 1 \le 2^k$ for integer $k \ge 1$, proved by induction. At $t = t_*$ the ratio is at most $1/3$, so the sum is at most $1/2$. Thus $1/2 \le q_{\kappa_e t_*} \le 3/2$. The free generator is the sum of commuting edge generators on the product space, so its kernel is the product of these N kernels. This proves (29). □

Lemma 7. *For every continuous $f \geq 0$ and $t \geq 0$,*

$$e^{-Mt} P_t f \leq e^{-tH} f \leq P_t f. \tag{32}$$

Consequently the exact vacuum satisfies

$$\frac{\max \psi_0}{\min \psi_0} \leq 3^N e^{Mt_*}, \qquad \frac{\min \rho}{\max \rho} \geq a_*, \qquad a_* \leq \rho(U) \leq a_*^{-1} \quad (U \in \mathcal{Q}). \tag{33}$$

Proof. Write $\mathcal{A} = \sum_e \kappa_e \Delta_e^T$, so $P_t = e^{t\mathcal{A}}$. The parabolic maximum principle on this compact manifold says that a solution of $(\partial_t - \mathcal{A} + V)u \geq 0$ with nonnegative initial data is nonnegative when $V \geq 0$. One direct proof replaces u by $e^{-ct}u + \epsilon$ with $c > 0$, considers a first negative minimum, and uses that $\partial_t u \leq 0$, $\mathcal{A}u \geq 0$ at that point; the positive zero-order coefficient makes the differential inequality impossible. Letting $\epsilon \downarrow 0$ proves the result. Smooth approximation covers continuous initial data. Apply this principle first to $u = e^{-tH} f$ to get $u \geq 0$. The free solution $v = P_t f$ is also nonnegative by the same argument with $V = 0$. Now $(\partial_t - \mathcal{A} + V)v = Vv \geq 0$, so $v - u \geq 0$. For $w = e^{-Mt}v$ the corresponding expression is $(V - M)w \leq 0$, so $u - w \geq 0$. This proves (32) with no approximation of V.

Apply it to $f = \psi_0$ and use $e^{-tH}\psi_0 = e^{-tE_0}\psi_0$. At time t_*, letting $I_0 = \int \psi_0 d\lambda > 0$, Lemma 6 gives

$$e^{(E_0 - M)t_*} 2^{-N} I_0 \leq \psi_0(U) \leq e^{E_0 t_*} (3/2)^N I_0.$$

Take the ratio of upper to lower bounds and square to obtain the first two assertions in (33). Since $\int \rho d\lambda = 1$, $\min \rho \leq 1 \leq \max \rho$. The ratio bound then gives $\min \rho \geq a_* \max \rho \geq a_*$ and $\max \rho \leq \min \rho / a_* \leq a_*^{-1}$, as required. □

Theorem 8. *For every simple edge loop,*

$$a_* \leq r_C \leq n^2 - a_*(n^2 - 1), \qquad a_*(n^2 - 1) \leq n^2 - r_C \leq n^2 - a_*. \tag{34}$$

For every such loop carrying a nontrivial center charge,

$$\frac{K_C}{2n} \frac{a_*(n^2 - 1)}{n^2 - a_*(n^2 - 1)} \leq Q_C \leq \frac{K_C}{2n} \frac{n^2 - a_*}{a_*}. \tag{35}$$

The full finite-lattice spectral gap, and also its gauge-invariant restriction above the unique vacuum, obey the genuine lower bound

$$\operatorname{gap}(H) \geq a_* \kappa_* \frac{n^2 - 1}{2n} > 0. \tag{36}$$

Proof. The density inequality $\rho \geq a_*$ and Lemma 4 give $r_C \geq a_*$ and $n^2 - r_C \geq a_*(n^2 - 1)$ by integrating the two nonnegative functions. These prove all of (34). For a charged loop the function $r \mapsto (n^2 - r)/r$ is strictly decreasing for $r > 0$; substitute the two endpoints into (26) to get (35).

For (36), first the exact free-product gap is $g_0 = \kappa_* C_F$. To verify the least nonzero single-edge Casimir, compute $(\omega_i, \omega_j) = \min(i, j) - ij/n > 0$ for $1 \leq i, j \leq n - 1$. If $m_i \geq 1$, all coefficients in $C(\lambda_m) - C(\omega_i)$ are nonnegative in the expansion in fundamental weights. Hence

$$C_m \geq C(\omega_i) = \frac{i(n - i)(n + 1)}{2n} \geq \frac{n^2 - 1}{2n}.$$

Equality occurs for a fundamental or antifundamental representation. On the product space each nonconstant matrix coefficient has at least one such nonzero edge term; taking a fundamental

coefficient on an edge with $\kappa_e = \kappa_*$ realizes g_0. Peter–Weyl expansion therefore proves the free Poincare inequality $\mathcal{E}_\lambda(f,f) \geq g_0 \operatorname{Var}_\lambda(f)$. For any complex f, the variational identity $\operatorname{Var}_\mu(f) = \min_{c\in\mathbb{C}} \int |f-c|^2 d\mu$ gives

$$\operatorname{Var}_\mu(f) \leq (\max \rho) \operatorname{Var}_\lambda(f), \qquad \mathcal{E}_\mu(f,f) \geq (\min \rho)\mathcal{E}_\lambda(f,f).$$

Consequently $\mathcal{E}_\mu(f,f) \geq [(\min\rho)/(\max\rho)] g_0 \operatorname{Var}_\mu(f) \geq a_* g_0 \operatorname{Var}_\mu(f)$. The exact unitary ground-state transform identifies this weighted Poincare infimum with the spectral gap of H. Restricting the infimum to gauge-invariant functions can only increase it, and the ground state belongs to that subspace by Lemma 2. □

The quotient bounds and the spectral bound have different logical roles: the upper bound on any one orthogonal trial quotient is an upper bound on the spectral gap, while a lower bound on that one quotient alone is not. Equation (36) is valid because its proof covers every form-domain vector. All constants in (28) retain volume and coupling dependence. For example, a_* decreases exponentially in N even with B fixed, and also decreases with Bt_*. Thus none of these inequalities asserts a strictly positive lower bound after removing a cutoff or taking infinite volume, and none proves that the limiting gap vanishes.

For a normalization check only, if V is identically zero then $\psi_0 = 1$, $r_C = 1$, and $m_C = 0$ for a simple loop, yielding $Q_C = K_C C_F$. This is an exact special case of the formula, not a replacement for the positive-coupling vacuum. The stated interacting results above apply with all $b_p > 0$.

7 Disjoint parallel plaquettes, covariance Fourier sums, and spectral weight

Let p_j, $j = 0, \ldots, m-1$, be simple plaquettes whose physical-edge supports are pairwise disjoint. Define

$$W_j = W_{p_j}, \quad f_j = W_j - \mu(W_j), \quad c_j(k) = e^{ikj}, \quad F_k = \sum_{j=0}^{m-1} c_j(k) f_j.$$

The mean subtraction is necessary for center-neutral plaquettes. It gives $\mu(F_k) = 0$ for every real k without any translation assumption. For each edge at most one f_j has a nonzero derivative, so pointwise

$$\Gamma_\kappa(F_k,F_k) = \sum_j |c_j(k)|^2 \Gamma_\kappa(W_j,W_j) = \sum_j \frac{K_j}{2n}(n^2 - |W_j|^2), \quad K_j = \sum_{e\in p_j} \kappa_e. \tag{37}$$

In particular its exact vacuum expectation is independent of k:

$$E_k := \mu(\Gamma_\kappa(F_k,F_k)) = \sum_j \frac{K_j}{2n}(n^2 - r_j), \qquad r_j = \mu(|W_j|^2). \tag{38}$$

Edge disjointness says nothing about statistical independence in μ. The full variance is

$$D_k := \mu(|F_k|^2) = \sum_{j,l=0}^{m-1} e^{ik(j-l)} C_{jl}, \qquad C_{jl} = \mu\big[(W_j - \mu W_j)\overline{(W_l - \mu W_l)}\big]. \tag{39}$$

All plaquette interactions, including plaquettes meeting more than one of these supports or connecting them indirectly, remain in C_{jl}. The covariance matrix is Hermitian positive semidefinite

because its quadratic form is the displayed squared $L^2(\mu)$ norm. In fact $D_k > 0$: if F_k were constant, varying an edge unique to p_j would force W_j to be constant, which it is not. Positivity of ρ then gives strictly positive variance. The exact quotient is

$$Q_k = \frac{E_k}{D_k}. \tag{40}$$

Suppose now the specified finite lattice and its full lists of couplings are invariant under a translation τ of order m taking p_j to p_{j+1} cyclically with their chosen orientations. Such a translation preserves H and Haar measure and therefore ψ_0 and μ by the same uniqueness argument as for gauge transformations. There are then constants $K_j = K$, $r_j = r$ and a cyclic covariance

$$C(d) = \mu(f_{l+d}\overline{f_l}), \qquad d \in \mathbb{Z}/m\mathbb{Z},$$

independent of l. For the allowed cyclic wave numbers $k = 2\pi q/m$, $q \in \{0, \dots, m-1\}$, set $\widehat{C}(k) = \sum_{d=0}^{m-1} e^{ikd} C(d)$. Regrouping the full double sum in (39), with $j = l + d$, proves

$$D_k = m\widehat{C}(k), \qquad E_k = m\frac{K}{2n}(n^2 - r), \qquad Q_k = \frac{K(n^2 - r)}{2n\widehat{C}(k)}. \tag{41}$$

Here $\widehat{C}(k)$ is real and strictly positive because it equals D_k/m; $C(-d) = \overline{C(d)}$ also proves reality directly. There is no algebraic k^2 factor in this numerator. Obtaining a small quotient in this family requires the actual interacting ratio on the right of (41) to become small. An assertion that all covariances vanish, or that the denominator has prescribed small-k behavior, is not contained in edge disjointness.

For real plaquette energy observables $A_j = n - \operatorname{Re} W_j$, replace f_j by $A_j - \mu A_j$ and use the same coefficients. All support and covariance arguments remain exact, but the local numerator is

$$e_j^{\mathrm{real}} = \frac{K_j}{4}\left[n - \mu\big(\operatorname{Re}\operatorname{tr}(H_{p_j}^2)\big) - \frac{2}{n}\mu\big((\operatorname{Im} W_j)^2\big)\right] \tag{42}$$

by (18). Thus under the same cyclic invariance $Q_k = e^{\mathrm{real}}/\widehat{C}_A(k)$ with $\widehat{C}_A$ the full covariance transform of A_j. Replacing this local numerator by the complex Wilson numerator would lose terms unless equality is proved in the particular group and observable.

Finally let $\mathsf{P}_{H-E_0}(\cdot)$ be the spectral projection measure of the nonnegative self-adjoint operator $H - E_0$ on the invariant Hilbert space. The normalized state and its scalar spectral probability measure are

$$v_k = \frac{\psi_0 F_k}{\sqrt{D_k}}, \qquad \nu_k(A) = \langle v_k, \mathsf{P}_{H-E_0}(A) v_k \rangle. \tag{43}$$

The orthogonality $\langle \psi_0, v_k \rangle = 0$ and uniqueness of the vacuum imply $\nu_k(\{0\}) = 0$. The spectral theorem and the ground-state identity prove the exact interacting first moment

$$\int_{[0,\infty)} \omega \, d\nu_k(\omega) = Q_k = \frac{E_k}{D_k}. \tag{44}$$

For every $\epsilon > 0$, integrating $\epsilon \mathbf{1}_{[\epsilon,\infty)}(\omega) \le \omega$ gives

$$\nu_k([\epsilon, \infty)) \le Q_k/\epsilon.$$

This is the exact connection between the computed variation and low-energy spectral weight. It makes no unproved assertion about $\widehat{C}(k)$, a continuum Hamiltonian, an infinite-volume vacuum, or convergence of the measures ν_k.

8 Verification and scope

The companion script `check_wilson_exact.py` uses exact SymPy arithmetic. On 8 September 2026 it completed with 1,012 exact scalar assertions, together with exact matrix-entry assertions. It verifies the basis, Fierz and Casimir constants for SU(2), SU(3), SU(4); the factor-four metric dictionary; explicit noncommuting SU(3) positive and negative edge variations and their cross contractions; real trace energy; a repeated physical edge with opposite signs; and the one-coordinate product-rule identity underlying (21). The proofs above are for all $n \geq 2$ and are not inferred from those finite checks. The script does not compute ψ_0, the interacting covariance matrix, or a continuum spectrum.

All main identities here concern the exact finite interacting Hamiltonian (3). They give explicit trial vectors and their exact relation to its spectral measure, as well as a positive but nonuniform finite-volume spectral bound. These proved relations can be carried into a separate continuum or geometric construction only through a proved state-space and operator map with controlled limits; no such map is claimed to be established by the present finite-lattice calculation.

References

[1] F. Peter and H. Weyl, *Die Vollständigkeit der primitiven Darstellungen einer geschlossenen kontinuierlichen Gruppe*, Mathematische Annalen **97** (1927), 737–755. doi:10.1007/BF01447892. This is the foundational compact-group completeness theorem used to identify the complete free spectral resolution.

[2] A. P. Polychronakos and K. Sfetsos, *Composing arbitrarily many $SU(N)$ fundamentals*, Nuclear Physics B **994** (2023), 116314, arXiv:2305.19345. The row/weight dictionary and dimension/Casimir expressions are given in Sections 2 and 4, including equation (4.30). Our symbol n is that paper's N; the derivations and all bounds used here retain the specified anti-Hermitian basis convention.